

ASSIGNMENT 3

QUESTION 1

hostname	ipconfig	getmac	ping	arp
nbstate	route	path	pathping	netstat
tracert				

Practice the commands mentioned in the above table and fill in the blanks.

SOLUTION

Command	Working
ping	Command is one of the most often used networking utilities for detecting devices on a network and for troubleshooting network problems.
hostname	command that displays the hostname of your machine
ipconfig	frequently used utility that is used for finding network information about your local machine like ip addresses, dns addresses etc.
nbstat	diagnostic tool for troubleshooting netbios problems
netstat	used for displaying information about tcp and udp connections and ports.
getmac	command that shows the mac address of your network interfaces
arp	for showing the address resolution cache. this command must be used with a command line switch -a is the most common.
tracert	command prints the path. if all routers on the path are functional, this command prints the full path.
path	command specifies the location where ms-dos should look when it executes a command.
route	command allows you to make manual entries into the network routing tables.
pathping	after sending out packets from you to a given destination, it analyzes the route taken and computes packet loss on a per-hop basis.

Outputs

ipconfig

txt

```
1 Windows IP Configuration
2
3
4 Ethernet adapter Ethernet 2:
5
6     Connection-specific DNS Suffix  . :
7     Link-local IPv6 Address . . . . . : fe80::5ca6:b97c:1b7f:ce94%9
8     IPv4 Address. . . . . : 172.16.119.58
9     Subnet Mask . . . . . : 255.255.255.0
```

```
10 Default Gateway . . . . . : 172.16.119.1
11
12 Wireless LAN adapter Local Area Connection* 1:
13
14 Media State . . . . . : Media disconnected
15 Connection-specific DNS Suffix . :
16
17 Wireless LAN adapter Local Area Connection* 6:
18
19 Media State . . . . . : Media disconnected
20 Connection-specific DNS Suffix . :
21
22 Wireless LAN adapter Wi-Fi 2:
23
24 Media State . . . . . : Media disconnected
25 Connection-specific DNS Suffix . :
```

ping

txt

```
1 Pinging google.com [172.217.26.14] with 32 bytes of data:
2 Reply from 172.217.26.14: bytes=32 time=10ms TTL=117
3 Reply from 172.217.26.14: bytes=32 time=10ms TTL=117
4
5 Ping statistics for 172.217.26.14:
6     Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

hostname

txt

```
1 DESKTOP-IDSI54N
```

netstat

txt

```
1 Active Connections
2
3 Proto Local Address          Foreign Address         State
4 TCP    127.0.0.1:49674          DESKTOP-IDSI54N:49675  ESTABLISHED
5 TCP    127.0.0.1:49675          DESKTOP-IDSI54N:49674  ESTABLISHED
6 TCP    127.0.0.1:59005          DESKTOP-IDSI54N:59436  ESTABLISHED
7 TCP    127.0.0.1:59006          DESKTOP-IDSI54N:59008  ESTABLISHED
8 TCP    127.0.0.1:59008          DESKTOP-IDSI54N:59006  ESTABLISHED
9 TCP    127.0.0.1:59333          DESKTOP-IDSI54N:59005  TIME_WAIT
10 TCP    127.0.0.1:59372          DESKTOP-IDSI54N:59005  TIME_WAIT
11 TCP    127.0.0.1:59394          DESKTOP-IDSI54N:59005  TIME_WAIT
12 TCP    127.0.0.1:59419          DESKTOP-IDSI54N:59005  TIME_WAIT
13 TCP    127.0.0.1:59436          DESKTOP-IDSI54N:59005  ESTABLISHED
14 TCP    172.16.119.58:58959      a23-208-178-166:https  ESTABLISHED
```

getmac

txt

```
1 Physical Address    Transport Name
2 =====
3 D8-9C-67-6F-4E-1F   Media disconnected
4 F4-39-09-22-9E-A7   \Device\Tcpip_{65BECBCF-235F-424E-82A7-FE2902C3E542}
5
6 arp -a
7 Interface: 172.16.119.58 --- 0x9
```

8	Internet Address	Physical Address	Type
9	172.16.119.1	74-8e-f8-90-80-78	dynamic
10	172.16.119.21	f4-39-09-22-9e-68	dynamic
11	172.16.119.27	f4-39-09-22-9d-b8	dynamic
12	172.16.119.54	f4-39-09-22-9f-2a	dynamic
13	172.16.119.88	f4-39-09-22-9d-c9	dynamic
14	172.16.119.89	f4-39-09-22-9f-15	dynamic
15	172.16.119.90	f4-39-09-22-9e-08	dynamic
16	172.16.119.91	f4-39-09-22-9d-ac	dynamic
17	172.16.119.95	f4-39-09-22-9d-8e	dynamic
18	172.16.119.96	f4-39-09-22-9e-73	dynamic
19	172.16.119.101	f4-39-09-22-9d-ab	dynamic
20	172.16.119.102	f4-39-09-22-9d-cf	dynamic
21	172.16.119.105	f4-39-09-22-9d-d9	dynamic
22	172.16.119.255	ff-ff-ff-ff-ff-ff	static
23	224.0.0.2	01-00-5e-00-00-02	static
24	224.0.0.22	01-00-5e-00-00-16	static
25	224.0.0.251	01-00-5e-00-00-fb	static
26	224.0.0.252	01-00-5e-00-00-fc	static
27	239.255.250.250	01-00-5e-7f-fa-fa	static
28	239.255.255.250	01-00-5e-7f-ff-fa	static

arp -a

txt

1	Interface: 172.16.119.58 --- 0x9		
2	Internet Address	Physical Address	Type
3	172.16.119.1	74-8e-f8-90-80-78	dynamic
4	172.16.119.21	f4-39-09-22-9e-68	dynamic
5	172.16.119.27	f4-39-09-22-9d-b8	dynamic
6	172.16.119.54	f4-39-09-22-9f-2a	dynamic
7	172.16.119.88	f4-39-09-22-9d-c9	dynamic
8	172.16.119.89	f4-39-09-22-9f-15	dynamic
9	172.16.119.90	f4-39-09-22-9e-08	dynamic
10	172.16.119.91	f4-39-09-22-9d-ac	dynamic
11	172.16.119.95	f4-39-09-22-9d-8e	dynamic
12	172.16.119.96	f4-39-09-22-9e-73	dynamic
13	172.16.119.101	f4-39-09-22-9d-ab	dynamic
14	172.16.119.102	f4-39-09-22-9d-cf	dynamic
15	172.16.119.105	f4-39-09-22-9d-d9	dynamic
16	172.16.119.255	ff-ff-ff-ff-ff-ff	static
17	224.0.0.2	01-00-5e-00-00-02	static
18	224.0.0.22	01-00-5e-00-00-16	static
19	224.0.0.251	01-00-5e-00-00-fb	static
20	224.0.0.252	01-00-5e-00-00-fc	static
21	239.255.250.250	01-00-5e-7f-fa-fa	static
22	239.255.255.250	01-00-5e-7f-ff-fa	static

tracert -h 5 google.com

txt

1	Tracing route to google.com [142.251.221.238]				
2	over a maximum of 5 hops:				
3					
4	1	<1 ms	2 ms	1 ms	172.16.119.1
5	2	<1 ms	<1 ms	<1 ms	172.31.1.6
6	3	20 ms	25 ms	36 ms	112.196.126.1

```
7 4 5 ms 4 ms 4 ms 202.164.32.25
8 5 4 ms 4 ms 4 ms 202.164.51.36
9
10 Trace complete.
```

path

txt

```
1 PATH=C:\ProgramData\Oracle\Java\javapath;C:\Program Files (x86)\Intel\Intel(R)
Management Engine Components\iCLS\;C:\Program Files\Intel\Intel(R) Management
Engine Components\iCLS\;C:\WINDOWS\system32;C:\WINDOWS;C:
\WINDOWS\System32\Wbem;C:\WINDOWS\System32\WindowsPowerShell\v1.0\;C:
\WINDOWS\System32\OpenSSH\;C:\Program Files (x86)\Intel\Intel(R) Management
Engine Components\DAL;C:\Program Files\Intel\Intel(R) Management Engine
Components\DAL;C:\Program Files (x86)\Intel\Intel(R) Management Engine
Components\IPT;C:\Program Files\Intel\Intel(R) Management Engine
Components\IPT;C:\Program Files\dotnet\;C:
\Users\CSED\AppData\Local\Microsoft\WindowsApps;
```

routeprint

txt

```
1 =====
2 Interface List
3 9...f4 39 09 22 9e a7 .....Intel(R) Ethernet Connection (7) I219-LM #2
4 17...da 9c 67 6f 4e 1f .....Microsoft Wi-Fi Direct Virtual Adapter #5
5 10...fa 9c 67 6f 4e 1f .....Microsoft Wi-Fi Direct Virtual Adapter #6
6 1.....Software Loopback Interface 1
7 14...d8 9c 67 6f 4e 1f .....Realtek RTL8821CE 802.11ac PCIe Adapter #2
8 =====
9
10 IPv4 Route Table
11 =====
12 Active Routes:
13 Network Destination Netmask Gateway Interface Metric
14 0.0.0.0 0.0.0.0 172.16.119.1 172.16.119.58 281
15 127.0.0.0 255.0.0.0 On-link 127.0.0.1 331
16 127.0.0.1 255.255.255.255 On-link 127.0.0.1 331
17 127.255.255.255 255.255.255.255 On-link 127.0.0.1 331
18 172.16.119.0 255.255.255.0 On-link 172.16.119.58 281
19 172.16.119.58 255.255.255.255 On-link 172.16.119.58 281
20 172.16.119.255 255.255.255.255 On-link 172.16.119.58 281
21 224.0.0.0 240.0.0.0 On-link 127.0.0.1 331
22 224.0.0.0 240.0.0.0 On-link 172.16.119.58 281
23 255.255.255.255 255.255.255.255 On-link 127.0.0.1 331
24 255.255.255.255 255.255.255.255 On-link 172.16.119.58 281
25 =====
26 Persistent Routes:
27 Network Address Netmask Gateway Address Metric
28 0.0.0.0 0.0.0.0 172.16.119.1 Default
29 =====
30
31 IPv6 Route Table
32 =====
33 Active Routes:
34 If Metric Network Destination Gateway
35 1 331 ::1/128 On-link
36 9 281 fe80::/64 On-link
```

```
37  9    281 fe80::5ca6:b97c:1b7f:ce94/128
38                                     On-link
39  1    331 ff00::/8                    On-link
40  9    281 ff00::/8                    On-link
41 =====
42 Persistent Routes:
43  None
```